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IMPLEMENTING A* ALGORITHM

Python Code:

```
from queue import PriorityQueue


def a_star(graph, start, goal, heuristic):

    open_set = PriorityQueue()
    open_set.put((0, start))

    came_from = {}

    g_score = {node: float('inf') for node in graph}
    g_score[start] = 0

    while not open_set.empty():

        _, current = open_set.get()

        if current == goal:
```

```
        path = []

        while current in came_from:
            path.append(current)
            current = came_from[current]

        path.append(start)

        return path[::-1]

    for neighbor, cost in graph[current]:

        tentative_g = g_score[current] + cost

        if tentative_g < g_score[neighbor]:

            came_from[neighbor] = current
            g_score[neighbor] = tentative_g

            f = tentative_g + heuristic[neighbor]

            open_set.put((f, neighbor))

return None
```

```
graph = {
    'A': [('B', 1), ('C', 4)],
    'B': [('D', 2), ('E', 5)],
    'C': [('F', 1)],
    'D': [('G', 2)],
    'E': [('G', 1)],
    'F': [('G', 3)],
    'G': []
}

heuristic =
    { 'A': 7,
    'B': 6,
    'C': 2,
    'D': 1,
    'E': 0,
    'F': 3,
    'G': 0
}

path = a_star(graph, 'A', 'G', heuristic)

print("Shortest path using A*:", path)
```

Output:

Shortest path using A*: ['A', 'B', 'D', 'G']

Result:

Thus the program of using the MINIMAX algorithm to evaluate the optimal move in a game tree was executed successfully.